A first-order primal-dual algorithm for convex problems with applications to imaging

(Chambolle, Pock, 2011)

(CP)

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Paper notes :-

Intro :-

↳ 1st order method (primal-dual) for convex problems with known saddle-point structure.

↳ $O(1/N)$ convergence :- non-smooth, finite dim

↳ $O(1/N^2)$ convergence :- primal / dual is uniformly convex

↳ $O(1/e^N)$ convergence :- both primal & dual are uniformly convex.

Pdhim :-

↳ $O(1/e^N)$ → linear convergence (ie. linear in log(.))
exponential in error

↳ Uniformly convex f^n

↳ General case of strongly convex f^n.

$$f(y) \geq f(x) + \nabla f(x)^T (y-x) + \phi(\|y-x\|)$$

where $\phi : [0, \infty) \to [0, \infty)$

($\phi(t) > 0$ for $t > 0$)

↳ Smoothness :-

↳ L-smooth for differentiable f if

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x-y\|$$

(gradient is Lipschitz)

↳ Useful in image problems

↳ denoising, deconvolution, inpainting, motion estimation, segmentation

↳ Motivation :- Many ill-posed ~~prob~~ inverse imaging problem

can be solved via minimizing → total variation
(useful for denoising; smoothing blurs edges also
TV preserves them)

$$\min_u \frac{1}{2}\|u-f\|_2^2 + \lambda \|\nabla u\|_i \quad (i = 1 \text{ or } 2)$$

↳ $i=1$ → makes it a good result but non-smooth f^n
↓
"cp tries to solve this"

The general problem

↳ X, Y → finite dim
↳ $K : X \to Y$: linear, continuous
↳ $\|K\| = \max_{\|x\| \le 1} \|Kx\|$

↳ Saddle-point problem :-

$$\min_{x \in X} \max_{y \in Y} \{ \langle Kx, y \rangle + G(x) - F^*(y) \}$$

(F^* : convex conjugate (later))

↳ $(\hat{x}, \hat{y})$ → saddle point if

$$L(\hat{x}, y) \le L(\hat{x}, \hat{y}) \le L(x, \hat{y})$$

where $L(x,y) = \langle Kx, y \rangle + G(x) - F^*(y)$

↳ $G : X \to [0, \infty)$; $F^* : Y \to [0, \infty)$
↳ proper, convex, lower semi-continuous (lsc)

Q) Pf prelim :-

↳ Proper f^n :- f is proper if it never takes $-\infty$ value & not identically $+\infty$.

↳ lsc → $f(x) \le \liminf_{n \to \infty} f(x_n)$ when $x_n \to x$

Q

↳ Why?
  ↳ Proper ensures finite minima exists.
  ↳ lsc ensures minima found from converging sequence is minima and the point has lower value (but not higher).
↳ Mapping to $[0, \infty)$ is convenience. Can shift additively to enable this without changing problem)

↳ This comes from primal problem

$$\min_{x \in X} \; f(Kx) + G(x)$$

(follows from TV problem str too) (many problems follow this too)

↓ separate $F(Kx)$ to F & K to make life easy.

Use convex conjugate

$$F(z) = \sup_y \{ \langle z, y \rangle - F^*(y) \}$$

$$\min_{x \in X} \max_{y \in Y} \{ \langle Kx, y \rangle - F^*(y) + G(x) \}$$ ← convex conjugate

→ saddle point problem

↳ $\langle Kx, y \rangle = \langle x, K^* y \rangle$ → adjoint

↳ Dual problem:-

$$\max_y \{ -G^*(-K^* y) - F^*(y) \}$$ ← useful to find optimality gap

↳ optimality condition:-

$-K\hat{x} \in \partial F^*(\hat{y})$

$-(K^* \hat{y}) \in \partial G(\hat{x})$ ← subgradient b/c f may be non-smooth

$D(y) \le p^* \le p(x)$

↳ $p^* - d^*$ is the duality gap.

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↳ Efficient when resolvent operators can be computed easily : (F & G are simple)

↳ Prelim :

↳ Resolvent operator :-

↳ $J_{\tau,\Lambda} = (I + \tau A)^{-1}$

How, instead of $\partial F(x)$ (say) [hard]

do $y - x \in \tau \partial F(x) \rightarrow \left(\frac{y-x}{\tau}\right) \in \partial F(x)$ and

$0 \in \partial F(x)$ ⇔ $(x \hookrightarrow y)$

Here, $x = (I + \tau \partial F)^{-1}(y)$

$\equiv$

$\arg\min_x \left\{ F(x) + \frac{1}{2\tau}\|x - y\|^2 \right\} \rightarrow (I + \tau \partial F)^{-1} = \text{prox}_{\tau F}$

Algorithm

↳ $y^{n+1} = \text{prox}_{\sigma F^*}(y^n + \sigma K \bar{x}^n)$ ← steps come from optimality conditions

$x^{n+1} = \text{prox}_{\tau G}(x^n - \tau K^* y^{n+1})$ ← the prox to find closest value

$\bar{x}^{n+1} = x^{n+1} + \theta(x^{n+1} - x^n)$

Init: Choose $\tau, \sigma > 0$, $\theta \in [0,1]$; $x^0, y^0 \in X \times Y$

$\bar{x}^0 = x^0$

↳ Since x, y coupled in saddle point ~~exp~~ extrapolation helps better convergence

$\theta = 1$ has nice math

↳ Step size condition for convergence → $\tau\sigma < \frac{1}{\|K\|^2}$

↳ Partial primal dual gap :-

$$G_{B_1 \times B_2}(x,y) = \max_{y' \in B_2} \langle y', Kx \rangle - F^*(y') + G(x) - \min_{x' \in B_1} \langle y, Kx' \rangle - F^*(y) + G(x')$$

$$\left\{ \max_{y' \in B_2} L(x,y') - \min_{x' \in B_1} L(x',y) \right\}$$

↳ we need bounded B_1 / B_2 set. (to ensure finite val)

(Theorem)

↳ Assume | saddle point exists.

→ $\theta = 1$

↳ $\tau \sigma \|L\|_2^2 < 1$

① Iterates are bounded

$$\frac{\|y^n - \hat{y}\|^2}{2\sigma} + \frac{\|x^n - \hat{x}\|^2}{2\tau} \leq C \left(\frac{\|y^0 - \hat{y}\|^2}{2\sigma} + \frac{\|x^0 - \hat{x}\|^2}{2\tau} \right)$$

where $C \leq \frac{1}{1 - \tau\sigma L^2}$

(Ensures Stability)

② The averaged iterates converge at $O(1/N)$

$$x_N = \frac{1}{N} \sum_{n=1}^{N} x^n \quad ; \quad y_N = \frac{1}{N} \sum_{n=1}^{N} y^n$$

↳ $G_{B_1 \times B_2}(x_N, y_N) \leq \frac{D(B_1, B_2)}{N}$

where $D(B_1, B_2) = \sup_{(x,y) \in (B_1 \times B_2)} \left\{ \frac{\|x - x^0\|^2}{2\tau} + \frac{\|y - y^0\|^2}{2\sigma} \right\}$

(Use averages s/c iterates might oscillate in saddle point problems)

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↳ Weak cluster points of (x_N, y_N) are saddle points (Limit of subsequence in infinite dim convergence is weak)

(3) In finite dim, actual iterates converge. //

↳ $x^k \to x^*$, $y^k \to y^*$ (x^*, y^*) → saddle point.

(But don't know explicit convergence rate)

↳ Proof intuition

↳ $0 < 1 - \tau\sigma\|K\|^2 < 1$ allows to control coupling term.

Proof shows all (3) via iterates telescoping series

Remark 1: When $\tau\sigma L^2 = 1$ → Control of both variables is lost (can show for one ordered in pair and a dual separately → in implementation, $\tau\sigma k^2 = 1 - \varepsilon$ for small $\varepsilon > 0$)

Remark 2: $O(1/N)$ only shown for averaged iterates gap restricted to $B_1 \times B_2$. For global, require more assumptions → show over $X \times Y$.

↳ Connections to existing work

If $\theta = 0$ → Arrow Hurwicz → $O(1/\sqrt{N})$ when Domain of F^* is bounded (or G is uniformly convex

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↳ Connection to extragradient method:-

↳ They show oscillations without extrapolation but employ a Prediction-correction. CP uses simple extrapolation inspired from this but approximate for efficiency.

↳ Connection to Douglas-Rachford splitting

↳ When $k = I$, it becomes theta splitting method and found a dual for easy optimization.

↳ Connection to preconditioned ADMM:-

↳ When done preconditioning on ADMM difficult terms cancel & becomes equivalent to CP.

(Alternate interpretation of CP)

Acceleration under cases

↳ If either is uniformly convex, update step sizes τ, θ, σ adaptively every iteration.

$$\theta_n = \frac{1}{\sqrt{1+2\gamma\tau_n}} \quad ; \tau_{n+1} = \theta_n \tau_n \quad , \sigma_{n+1} = \frac{\sigma_n}{\theta_n}$$

& $O(\frac{1}{N^2})$ convergence.

↳ $\tau = \frac{1}{L}\sqrt{\frac{\delta}{\gamma}}$, $\sigma = \frac{1}{L}\sqrt{\frac{\gamma}{\delta}}$; $L = \|k\|$

$$\theta \in \left[\frac{1}{1+\mu}, 1\right] \quad , \mu = \frac{2\sqrt{\gamma\delta}}{L}$$

↳ Linear convergence, $O(\frac{1}{e^N})$

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→ Applications

↳ ROF model

↳ Works well for Gaussian noise but not impulse noise. (salt-pepper)

↳ L2 norm for data fidelity.

↳ $\min_u \|\nabla u\|_1 + \frac{\lambda}{2}\|u - g\|_2^2$

↳ Proximal maps are cheap → P^* → Projection on unit ball

↳ G → Can be solved!

↳ $p_{ij}^{k+1} = \frac{\tilde{p}_{ij}}{\max(1, |\tilde{p}_{ij}|_2)}$; $u_{ij}^{k+1} = \frac{\tilde{u}_{ij} + \tau\lambda g_{ij}}{1 + \tau\lambda}$

↳ G is L2 norm → smooth ⇒ $O(1/N^2)$ convergence!

→ Under strong TV → May have unwanted flat regions

↳ Hair staircasing.

↳ Makes shrinkage harder.

↳ b/c $\nabla = 0$ (strong TV)

@ $\nabla = 0$ → non-differentiable. Thus harder.

↳ Not good for salt-pepper noise.

↳ TV-L^1 model

↳ Replace data fidelity term with L1 norm.

→ Robust to outliers → salt-pepper noise.

→ TV-L^1 is contrast invariant.

↳ Since both terms L1; can take out equal scaling as common. (in ROF they scale different. So affects denoising)

↳ Huber - ROF

↳ $\min_u \sum \phi_\alpha(\|(\nabla u)_{ij}\|_2) + \frac{\lambda}{2}\|u-g\|_2^2$

↳ $\phi_\alpha(t) = \begin{cases} t^2/2\alpha & \cdot |t| \le \alpha \\ |t| - \frac{\alpha}{2} & \cdot |t| > \alpha. \end{cases}$

Here, $F^k(p) = f_p(p) + \frac{\alpha}{2}\|p\|_2^2$.

↳ Reduces staircasing effect. in domain (i.e. Ω) → strongly convex. → Thus $O(1/k^2)$ convergence

→ Thus better than ROF (for staircasing but still ~~does~~ is sive with large noise.)

↳ Advanced Applications

1) Image deconvolution & zooming

↳ Deconvolution :- Remove Blur

→ Zooming :- low-resolution image → high-resolution

$\min_u TV(u) + \frac{\lambda}{2}\|Au - g\|_2^2$

(A: Degradation operator)

↳ When A is convolution → proximal map is efficiently computed via FFT.

2) Image inpainting

↳ Recover corrupted portions of image.

$\min_u \|\phi u\|_1 + \frac{\lambda}{2} \cdot \sum_{(i,j) \in D-I} (u_{ij} - g_{ij})^2$

($\|\phi u\|_1$: Regularization prior.)

∴ can be gradient ROF further.

↳ 3) Motion Estimation (optical flow)

↳ TV regularization for smooth motion fields

• Fidelity is brightness constancy assumption.

→ 4) Image segmentation.

↳ Minimize region boundary & related cost of label assignment.

Broadly,

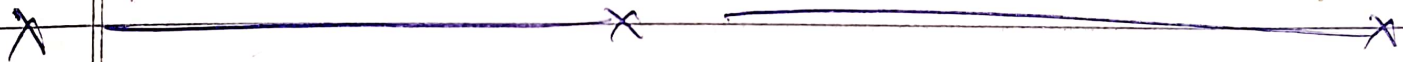
$\min_{\alpha} R(\text{term}) + R(\alpha)$ form !

(Ref paper for details)

↳ Limitations

↳ convex assumption, $\nabla^T \|k\|^2 < 1$ → can be ill-conditioned

↳ Dependence on λ.



Derivation for 3D point cloud color denoising :-

↳ Point cloud with N points : $\{p_1, \ldots, p_N\} = P$

$p_i \in \mathbb{R}^3$

↳ Graph G

↳ $V = p_i$, $E =$ K ~~Neighbour~~ nearest neighbours (symmetric)

$W : w_{ij} = \exp\left(-\frac{\|p_i - p_j\|_2^2}{h}\right)$ ↳ distance based weighting on edge

For each channel c,

$f^c = [f_1^c \ldots f_N^c]^T$; $u^c = [u_1^c \ldots u_N^c]^T$

↳ Graph gradient :-

$(\nabla u^c)_e = \sqrt{w_{ij}} \cdot (u_j^c - u_i^c)$

$\therefore TV_G(u^c) = \sum_{(i,j) \in E} \sqrt{w_{ij}} \, |u_j^c - u_i^c| = \|\nabla u^c\|_1$

↳ Edge-wise anisotropic graph TV.

$\therefore \min \underbrace{\|\nabla u^c\|_1}_{F(\nabla u)} + \underbrace{\frac{\lambda}{2}\|u^c - f^c\|_2^2}_{G(u)}$

↳ ∇^* derivation :

↳ Weighted graph divergence

$(\nabla^* q)_i = \sum_{e \in (j,i)} \sqrt{w_e}\, q_e - \sum_{e \in (i,j)} \sqrt{w_e}\, q_e$

In my code, I implement isotropic version.

(L2 over each vertices and then L1)

X ——— X ———→ X